

Speed limit

1. Course Content

Note: This program runs on a sandbox map; you will need to adjust the program for your personal map.

1. Learn the actions the car takes after recognizing the speed limit.
2. Learn the newly emerging key source code.

2. Speed Limit Detection

Speed limit detection source code path:

```
/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/yolo_detect.py
```

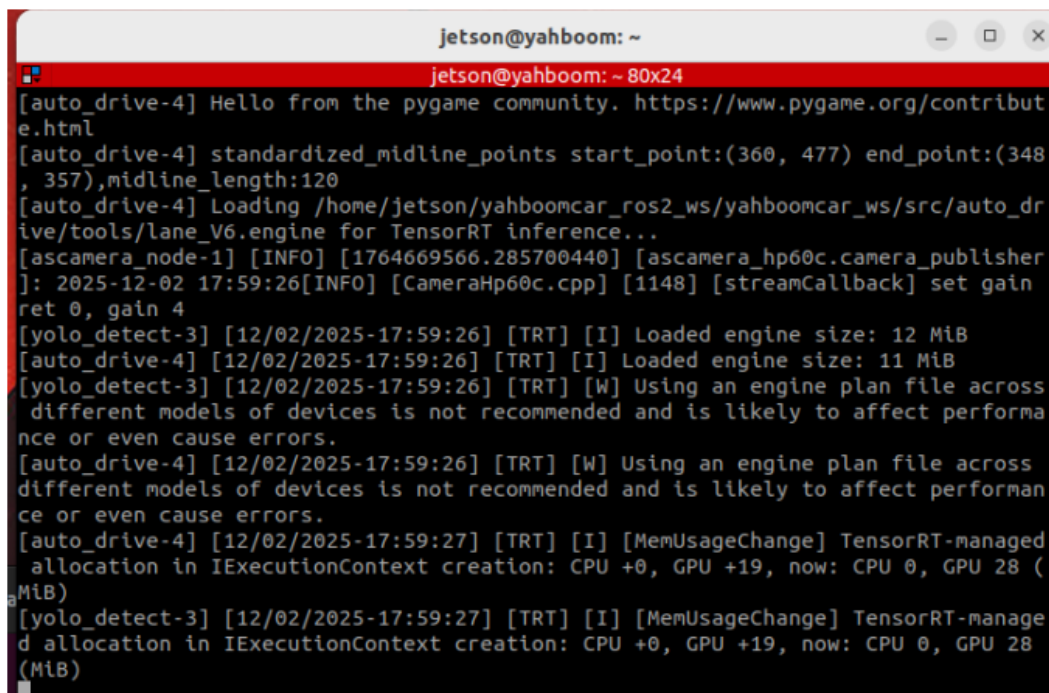
```
/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/auto_drive.py
```

3. Program Startup

3.1 Startup Command

- Start the camera + chassis + road signs, lane lines

```
ros2 launch auto_drive auto_drive_bringup.launch.py
```



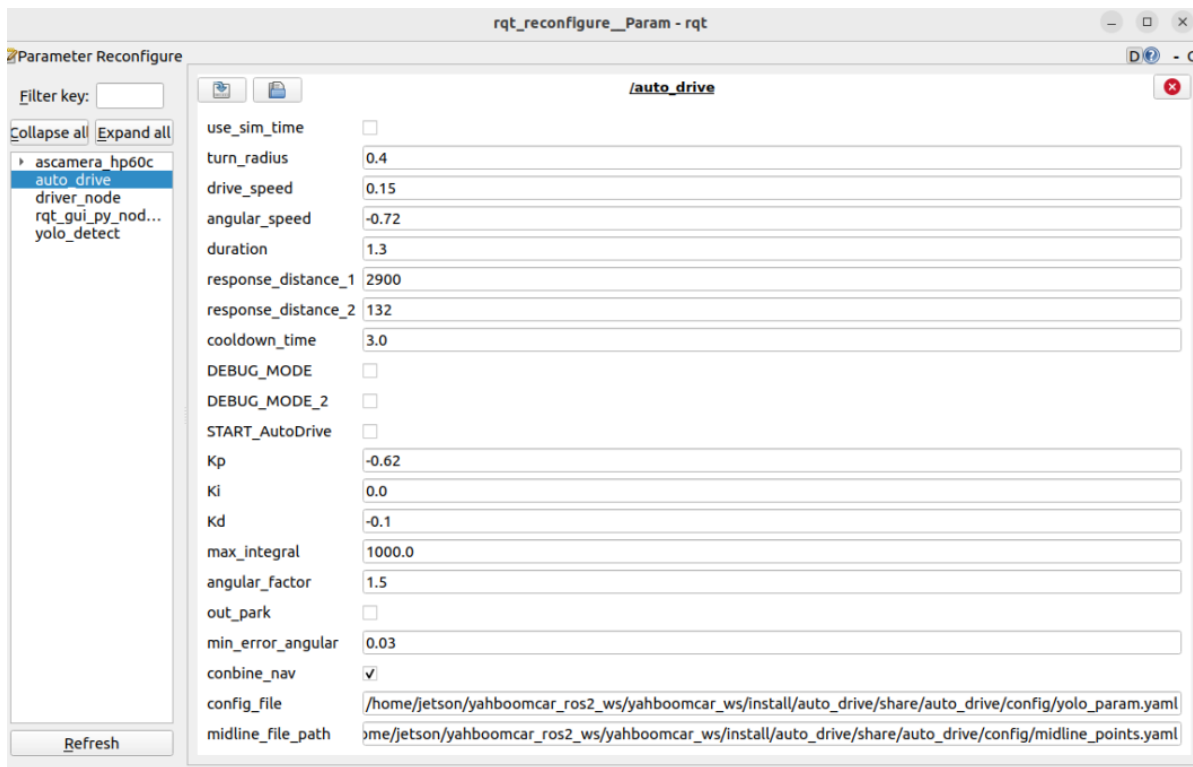
```
jetson@yahboom: ~  
jetson@yahboom: ~ 80x24  
[auto_drive-4] Hello from the pygame community. https://www.pygame.org/contribut  
e.html  
[auto_drive-4] standardized_midline_points start_point:(360, 477) end_point:(348  
, 357),midline_length:120  
[auto_drive-4] Loading /home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_dr  
ive/tools/lane_V6.engine for TensorRT inference...  
[ascamera_node-1] [INFO] [1764669566.285700440] [ascamera_hp60c.camera_publisher  
]: 2025-12-02 17:59:26[INFO] [CameraHp60c.cpp] [1148] [streamCallback] set gain  
ret 0, gain 4  
[yolo_detect-3] [12/02/2025-17:59:26] [TRT] [I] Loaded engine size: 12 MiB  
[auto_drive-4] [12/02/2025-17:59:26] [TRT] [I] Loaded engine size: 11 MiB  
[yolo_detect-3] [12/02/2025-17:59:26] [TRT] [W] Using an engine plan file across  
different models of devices is not recommended and is likely to affect performa  
nce or even cause errors.  
[auto_drive-4] [12/02/2025-17:59:26] [TRT] [W] Using an engine plan file across  
different models of devices is not recommended and is likely to affect performan  
ce or even cause errors.  
[auto_drive-4] [12/02/2025-17:59:27] [TRT] [I] [MemUsageChange] TensorRT-manage  
d allocation in IExecutionContext creation: CPU +0, GPU +19, now: CPU 0, GPU 28 (MiB)  
[yolo_detect-3] [12/02/2025-17:59:27] [TRT] [I] [MemUsageChange] TensorRT-manage  
d allocation in IExecutionContext creation: CPU +0, GPU +19, now: CPU 0, GPU 28 (MiB)
```

After the program runs, a YOLO recognition screen will be displayed. The car will move centered within the lane and perform corresponding actions based on the road signs it detects during its journey. (If the terminal does not report any errors after starting the program, but the car does not move, you can run the following program to check the parameters.)



- Start the dynamic parameter configuration node

```
ros2 run rqt_reconfigure rqt_reconfigure
```



If the car doesn't move, check two parameters: whether `START_AutoDrive` is checked and whether `combine_nav` is false.

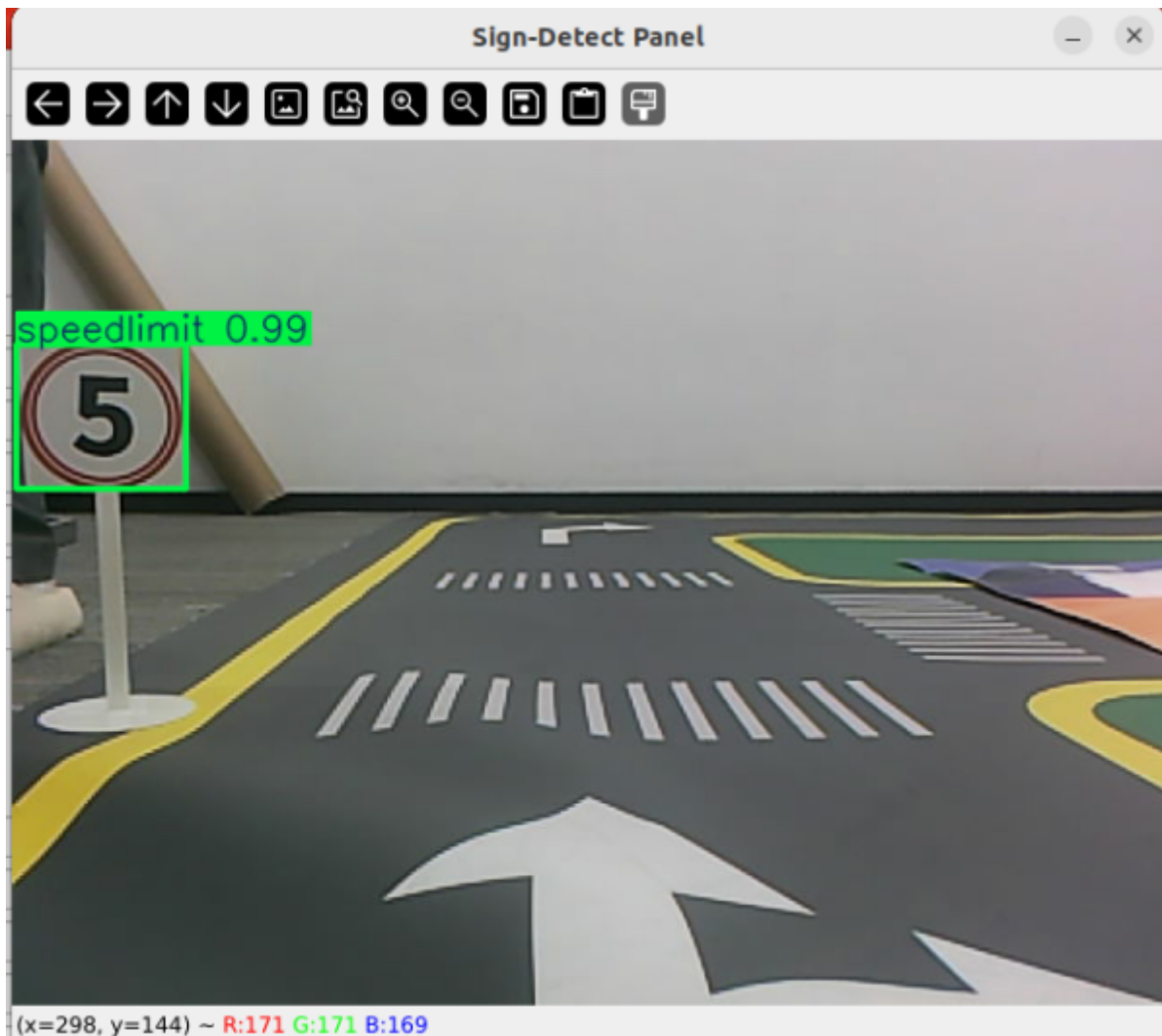
✓ After modifying the parameters, click in the blank area of the GUI to enter the parameter values. Note that this only applies to the current startup; for permanent effects, the parameters need to be modified in the source code.

3.2 Car Movement After Speed Limit Detection

The program will decelerate for 3 seconds after detecting a speed limit and then return to normal.

```
jetson@yahboom: ~
jetson@yahboom: ~ 80x24
[auto_drive-4] [INFO] [1764671378.387579116] [auto_drive]: current_sign redlight
[auto_drive-4] [INFO] [1764671378.390984163] [auto_drive]: red_stop
[auto_drive-4] [INFO] [1764671832.607130196] [auto_drive]: current_sign greenlight
[auto_drive-4] [INFO] [1764671832.608561506] [auto_drive]: greenlight
[auto_drive-4] [INFO] [1764673308.321102763] [auto_drive]: current_sign parkA
[auto_drive-4] [INFO] [1764673308.323742671] [auto_drive]: parking A
[auto_drive-4] [INFO] [1764674001.122264527] [auto_drive]: current_sign speedlimit
[auto_drive-4] [INFO] [1764674001.122991778] [auto_drive]: slow down
[auto_drive-4] [INFO] [1764674013.459037728] [auto_drive]: current_sign stop
[auto_drive-4] [INFO] [1764674016.463009335] [auto_drive]: Stop action completed
[auto_drive-4] [INFO] [1764674377.727182525] [auto_drive]: current_sign turnright
[auto_drive-4] [INFO] [1764674377.728323130] [auto_drive]: Turn action preparing
: right, waiting 2 seconds...
[auto_drive-4] [INFO] [1764674377.729364725] [auto_drive]: Turn action started:
right
[auto_drive-4] [INFO] [1764674378.732186552] [auto_drive]: turn right
[auto_drive-4] [INFO] [1764674618.914934213] [auto_drive]: current_sign honking
[auto_drive-4] [INFO] [1764674618.915651776] [auto_drive]: car horn
[auto_drive-4] [INFO] [1764674729.068016763] [auto_drive]: current_sign sidewalk
[auto_drive-4] [INFO] [1764674729.069482311] [auto_drive]: slow down
```

The car will decelerate for 3 seconds.



The fixed-point parking handling function is located in the `auto_drive.py` file. Below is the function entered after the speed limit is detected. Users can modify the car's movement after detecting a road sign. When in road network navigation mode, it will continuously issue deceleration commands.

```
def slow_down(self):
    '''减速 slow down'''
    self.get_logger().info('slow down')
    if self.enable_route_nav != True:
        self.drive_speed = 0.1
        time.sleep(3.0)
        self.drive_speed = 0.15
    else:
        self.current_speed.linear.x=0.05
        self.current_speed.angular.z=0.0
        self.cmd_vel_pub.publish(self.current_speed)#D挡,正常行驶
```

4. Source Code Analysis

4.1、auto_drive.py

YOLO signpost message reception:

```
def sign_callback(self, msg:DetectionObject):
    '''Subscribe to YOLO recognition results'''
    class_name = msg.class_name
    object_distance = self._get_distance(msg.ymin) # Calculate distance
    object = Object(msg.class_name, msg.confidence, object_distance)
```

Multiple filtering mechanisms: mainly to prevent continuously recognizing road signs and triggering actions; the action will only be unlocked when different road signs are recognized.

```
# 1. Check if any other actions are being performed.
if self.action_runing:
    return

# 2. Check cooldown time (to prevent repeated triggering)
current_time = time.time()
if class_name in self.last_processed_time:
    time_diff = current_time - self.last_processed_time[class_name]
    if time_diff < self.cooldown_time: # Default cooldown: 3 seconds
        return

# 3. Check if it is a duplicate road sign.
if class_name == self.last_sign:
    return

# 4. Check if the road sign is enabled.
if not self.sign_enable_list[对应的索引]:
    return
```

Event queue processing

```
def handle_events(self):
    '''Handling target recognition events'''
    while True:
        if not self.event_queue.empty():
            object = self.event_queue.get()
            self.action_runing = True
            current_sign = object.class_name

            try:
                # Perform the appropriate action based on the type of road sign.
                if object.class_name == "straight" and
self.sign_enable_list[10]:
                    self.go_straight()
                elif object.class_name == "turnright" and
self.sign_enable_list[11]:
                    self.turn('right')
                # ... Other road sign processing

            finally:
                # Update status record
                self.last_processed_time[current_sign] = time.time()
                self.last_sign = current_sign
                self.action_runing = False
```

Specific action implementation

```
def slow_down(self):
    '''减速 slow down'''
    self.get_logger().info('slow down')
    if self.enable_route_nav != True:
        self.drive_speed = 0.1
        time.sleep(3.0)
        self.drive_speed = 0.15
    else:
        self.current_speed.linear.x=0.05
        self.current_speed.angular.z=0.0
        self.cmd_vel_pub.publish(self.current_speed)#D挡, 正常行驶
```